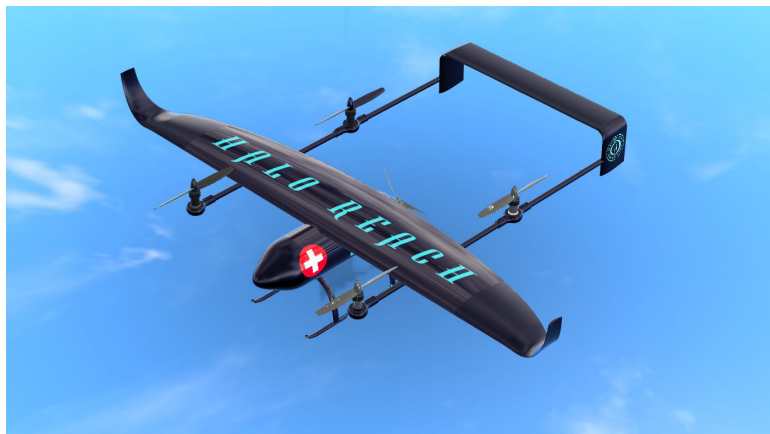

HALO REACH: Longitudinal Fixed Wing Control Design

UAS for Disaster Relief - AE3 2024



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Abstract

This report outlines the longitudinal state space analysis and development of airspeed and altitude controllers for the fixed wing configuration of the HALO REACH UAV. The controllers are designed to meet criteria outlined by the RPV Flying Qualities [1]. The airspeed controller consists of a single closed loop PID architecture, while the altitude controller encompasses a triple cascade PID structure. Robustness analysis is carried out against wind gusts and payload/fuel mass changes corresponding to the different mission profiles of the UAV. Ultimately both controllers were deemed to successfully track reference inputs, attenuate perturbations and adapt to varying mass configurations.

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Nomenclature

α	Angle of Attack
$\bar{c}$	Mean Aerodynamic Chord
δ_e	Elevator Deflection Angle
ϵ	Downwash Angle
γ	Climb Gradient
ω_n	Natural Frequency
ρ	Density
Θ	Pitch Angle
ζ	Damping Ratio
a_E	Elevator Lift Curve Slope
C_D	Drag Coefficient
C_L	Lift Coefficient
C_M	Pitching Moment Coefficient
C_T	Thrust Coefficient
d_E	Elevator Drag Curve Slope
h_H	Vertical Distance between Horizontal Tailplane and Centre of Gravity
K_D	PID Derivative Gain
K_I	PID Integral Gain
K_n	Static Margin
K_P	PID Proportional Gain
l_H	Horizontal Distance between Horizontal Tailplane and Centre of Gravity
l_T	Vertical distance between Propellor and Centre of Gravity
N	PID Filter Coefficient
q	Pitch Rate
S_H	Horizontal Tailplane Reference Area
S_{ref}	Wing Reference Area
T	Thrust
U	Longitudinal Velocity, Airspeed
U_e	Equilibrium Velocity
u_T	Thrust Control Input

u_{δ_e}	Elevator Deflection Control Input
V_∞	Freestream Velocity
W	Vertical Velocity
z	Altitude
HALO	Humanitarian Aid Logistics Operations
PID	Proportional, Integral, Derivative
REACH	Rapid Emergency Aid and Communications Hub
RPV	Remotely Piloted Vehicle
SPPO	Short Period Pitch Oscillation
UAS	Uncrewed Aerial System
UAV	Unmanned Aerial Vehicle
VTOL	Vertical Take-Off and Landing

1 Introduction

Disaster relief is an increasingly prevalent area of engineering focus and innovation. Communications breakdowns, infrastructure damage and harsh weather are some of the key obstacles faced by relief efforts in disaster zones such as the earthquake that devastated Turkey and parts of Syria in February 2023 [2]. Said obstacles adversely affect the usability of conventional transport modes such as ground vehicles and fixed wing aircraft, thereby creating scope for a Vertical Take-Off and Landing (VTOL) Unmanned Aerial Vehicle (UAV) capable of transporting relief supplies and serving as a long-endurance communications relay [2]. Flight dynamics and control analysis are key to developing such an aircraft, facilitating flight stability and robustness to harsh weather (particularly wind gusts). Controllers for this UAV encompass a large scope including controllers for VTOL, fixed wing cruise/loiter and transition between the two modes of flight. This report primarily focuses on analysis of the UAV's longitudinal state space and development of airspeed and altitude controllers for the fixed wing configuration. Robustness analysis is also carried out to ensure satisfactory performance under wind gusts and varying payload/fuel mass.

2 Control Aims

Prior to starting development of the controllers, it is important to define the aims and expectations from performance of the controllers. The RPV (Remotely Piloted Vehicle) flying qualities design criteria [1] can be found in Table 2. Despite these criteria stipulating a wind gust resistance of up to 15.2m/s, our competitors advertise a wind gust resistance of 18m/s [3] hence defining our new target. In addition zero steady state error and overshoot requirements were set for accuracy.

Table 2. RPV Control Design Criteria

Parameter	Design Criteria
Airspeed Accuracy	$\pm 5\text{kn}$
Altitude Accuracy	$\pm 30\text{ft}$
Gain Margin	$> 6\text{dB}$
Phase Margin	$> 45^\circ$
Max. Wind Gust Amplitude	18m/s
Steady State Error	0
Overshoot	0

In addition to these criteria, the controller must achieve satisfactory performance across all configurations of payload and fuel mass, corresponding to the different missions required of our UAV. These parameters are summarised in Table 3. The zero fuel, zero payload configuration is also included to represent phases of the mission following the payload drop and varying fuel mass due to consumption.

Table 3. Mass Configurations for each Mission

Mission	Payload Mass (kg)	Fuel Mass (kg)	Total Mass (kg)	Centre of Gravity (m)
Supply Delivery	50	6.5	160	1.15
Long Endurance	20	36.5	160	1.08
0 fuel, 0 Payload	0	0	103.5	1.20

The airspeed and altitude controllers must also minimise aggression. In the case of airspeed, an aggressive airspeed controller will adversely affect stability and trajectory tracking. Looking at altitude, an aggressive altitude controller may result in large climb gradients, affecting aerodynamic performance and negating the validity of our linearised state space about the cruise angle of attack.

3 Control Methodology

3.1 Longitudinal State Space

To provide a model to test and tune the controllers on, a state space encompassing the longitudinal physics of the UAV was developed. Considering only the symmetric states (longitudinal velocity U , vertical velocity W , pitch rate q & pitch angle Θ) the state space was formed using the UAV's stability derivatives, calculated from the equations outlined in Appendix A [4]. The state space was evaluated and linearised about the 500m altitude, 41.1m/s cruise equilibrium using mass parameters from the supply delivery mission (see Table 3).

As for the control derivatives, the longitudinal control system acts through two control inputs, u_{δ_e} and u_T , which respectively adjust the elevator deflection δ_e and thrust T . Table 4 illustrates the ranges of these parameters and their equilibrium values. The ranges of u_{δ_e} and u_T are designed to represent the fraction of max. δ_e and T respectively.

Table 4. Control Input Definition

Parameter	Min. Value	Max. Value	Equilibrium Value
δ_e	-30°	30°	0°
u_{δ_e}	-1	1	0
T	0N	320N	190N
u_T	-0.6	0.4	0

The control derivatives were evaluated about the same equilibrium and adjusted for the formatting of the control inputs u_{δ_e} and u_T . The full longitudinal state space equation can be seen in Equation (1).

$$\begin{bmatrix} \dot{U} \\ \dot{W} \\ \dot{q} \\ \dot{\Theta} \end{bmatrix} = \begin{bmatrix} -0.0607 & 0.0498 & -0.0593 & -9.8100 \\ -0.5046 & -3.6757 & 39.7264 & 0 \\ 0.0036 & -0.0570 & -0.9856 & 0 \\ 0 & 0 & 1.000 & 0 \end{bmatrix} \begin{bmatrix} U \\ W \\ q \\ \Theta \end{bmatrix} + \begin{bmatrix} -0.5722 & 2.0000 \\ -7.3634 & 0 \\ -7.2713 & 0.0376 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} u_{\delta_e} \\ u_T \end{bmatrix} \quad (1)$$

The longitudinal eigenvalues of the UAV are illustrated in Figure 3. As expected we observe two stable oscillatory modes. One with a higher frequency corresponding to the Short Period Pitch Oscillation (SPPO) mode and the other corresponding to the lower frequency Phugoid mode [4]. Analysing the natural frequency ω_n of each of these modes highlights frequencies of wind gusts that should be covered during robustness analysis in order to capture the largest state deviations.

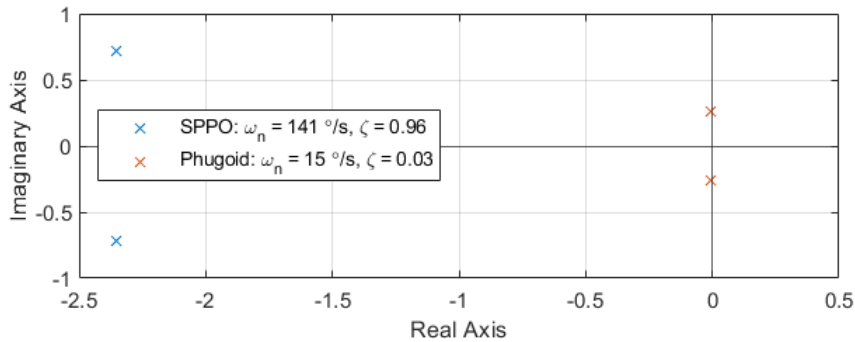


Figure 3. Longitudinal UAV Eigenvalues

3.2 Tuning Approach

PID control was chosen as the primary control method. Equation (2), showing the transfer function of a PID controller [5], illustrates how the gains can be selected to place two poles and two zeros allowing

for manipulation of the closed loop behaviour.

$$\frac{(K_P + K_D)s^2 + (K_P N + K_I)s + K_I N}{s(s + N)} \quad (2)$$

Figure 4 displays the tuning process employed for each of the controllers developed. The performance of each controller was primarily evaluated using step reference tracking, allowing for observation of compliance with the aims outlined in section 2. Appendix B displays the tuning environment used in MATLAB's Control System Designer App [6]. Upon completion the initial tuning procedure outlined, the controllers were verified and tested using Gazebo, a high fidelity simulator, the results of which are covered by Lee J. [7].

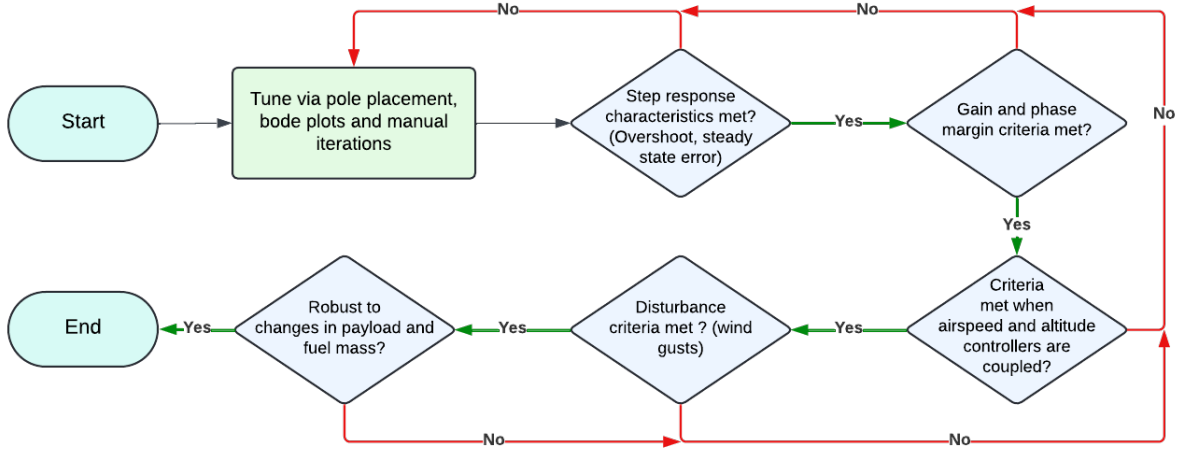


Figure 4. Tuning Process Flowchart

4 Airspeed Control

Prior to tuning the airspeed controller, the state space was simplified by isolating the states relevant to Phugoid and as seen in Equation (3). The SPPO mode only concerns the W and q states [4], hence is not considered for this controller. Regarding control inputs, only the u_T control input is used in this instance.

$$\begin{bmatrix} \dot{U} \\ \dot{W} \end{bmatrix} = \begin{bmatrix} -0.0607 & 0.0498 \\ -0.5046 & -3.6757 \end{bmatrix} \begin{bmatrix} U \\ W \end{bmatrix} + \begin{bmatrix} 2.0000 \\ 0 \end{bmatrix} [u_T] \quad (3)$$

By neglecting the q and Θ states, it's assumed that the elevator controller will be able to remove perturbations in these states. Since the SPPO mode has a much higher frequency than the Phugoid mode [4], this is a sensible assumption since an elevator controller capable of mitigating the SPPO mode will be adept at eliminating perturbations in the neglected states such that they can be decoupled from the U state.

The controller architecture used can be seen in Figure 5, employing a single PID closed loop structure to track reference airspeed. Following initial tuning, the controller performance can be observed in Figure 6 against both a unit step reference, and a more realistic step reference of 10m/s. The control output u_T is converted and shown as a percentage of maximum thrust for ease of interpretation. For the unit step reference, our controller achieves all stipulated goals outlined in Table 2. The 10m/s step caused the thrust output to saturate, likely due to a decrease in propeller performance with increased speed, however this did not affect the tracking significantly. The step response characteristics and

controller parameters can be seen in Table 5 illustrating how the rise and settle time were kept lenient in order to avoid having an excessively aggressive controller.

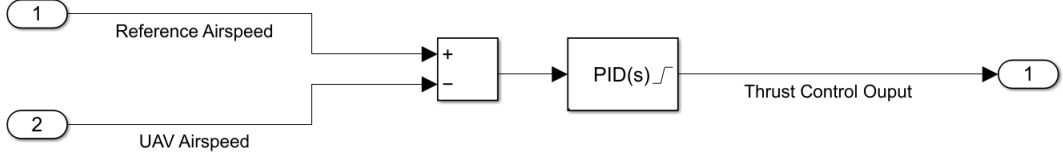


Figure 5. Airspeed Controller Architecture

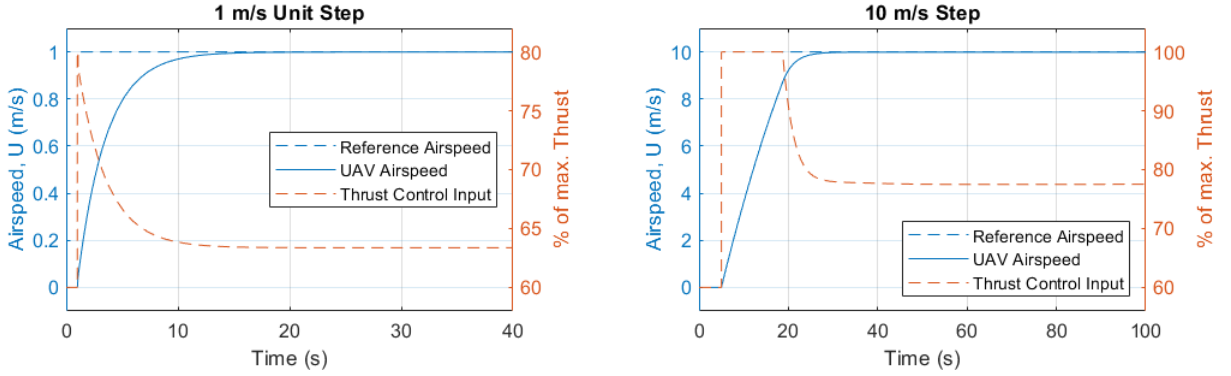


Figure 6. Airspeed Controller Step Reference Tracking

Table 5. Airspeed Unit Step Response Characteristics and Controller Parameters

Rise Time (s)	Settle Time (s)	Gain Margin (dB)	Phase Margin ($^{\circ}$)	K_P	K_I	K_D	N
6.33	17.53	∞	91.4	8.00	0.01	20.00	100

5 Altitude Control

Initially an identical control architecture as seen in Figure 5 was used to control the altitude z via elevator input, however this resulted in excessively oscillatory and inaccurate responses. Altitude change is coupled with changing the pitching angle Θ /climb gradient γ and subsequently the pitch rate q . The absence of control over the coupled states can lead to unmitigated excitation of the SPPO mode resulting in the aforementioned oscillations and inaccuracies in the altitude response. Consequently, a triple cascaded PID control structure as seen in Figure 7 was employed [8].

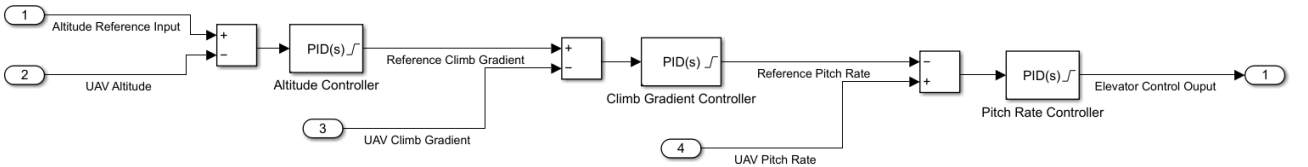


Figure 7. Altitude Controller Architecture

This control structure consists of an inner pitch rate loop, middle climb gradient loop and outer altitude loop. The ordering of the loop was chosen to place states that change with the highest

frequency inwards. This is done such that perturbations in the inner states can be rapidly mitigated with minimal disturbance to tracking of the outer state [8]. From a step response perspective, this results in shorter rise time requirements for the inner loop controllers and more lenient rise times for the outer controllers. The PID controllers in this system are tuned from the inside out. Successive PID controllers are tuned assuming perfect control of the inner controllers over their respective states [8].

5.1 Pitch Rate Controller

The pitch rate controller was tuned on a state space isolating the states relevant to SPPO as seen in Equation (4) [4].

$$\begin{bmatrix} \dot{W} \\ \dot{q} \end{bmatrix} = \begin{bmatrix} -3.6757 & 39.7264 \\ -0.0570 & -0.9856 \end{bmatrix} \begin{bmatrix} W \\ q \end{bmatrix} + \begin{bmatrix} -7.3634 \\ -7.2713 \end{bmatrix} [u_{\delta_e}] \quad (4)$$

Similar to before, it is assumed in this case that the airspeed controller is sufficiently able to maintain constant longitudinal velocity U facilitating neglect of this state. Changes in weight forces due to the pitch angle Θ can be considered negligible compared to aerodynamic lift and pitching moment effects [4] meaning this state can be omitted. The unit step response tracking of the tuned pitch rate controller can be seen in Figure 8.

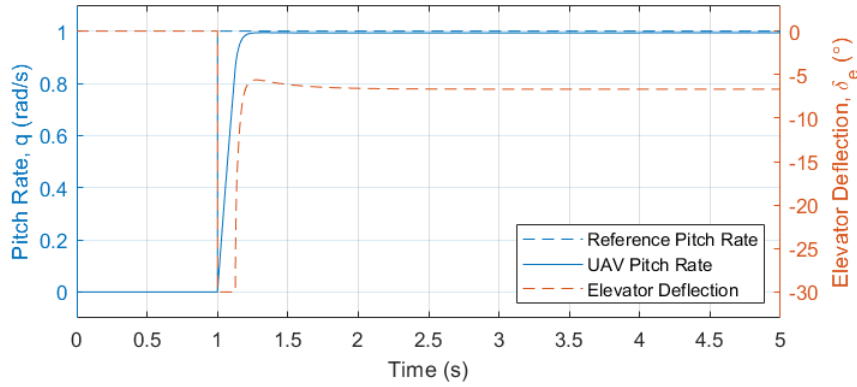


Figure 8. Pitch Rate Controller Step Reference Tracking

The pitch rate controller was tuned to have a minimal rise time due to reasons outlined earlier resulting a rapid step response. The controller was finalised after ensuring compliance with the criteria outlined in Table 2 and the unit step response characteristics and control parameters can be seen in Table 6.

Table 6. Pitch Rate Unit Step Response Characteristics and Controller Parameters

Rise Time (s)	Settle Time (s)	Gain Margin (dB)	Phase Margin (°)	K_P	K_I	K_D	N
0.14	0.31	∞	89.3	20.00	1.00	0.30	100

5.2 Climb Gradient Controller

The climb gradient controller is the next controller in the cascade loop, and it commands a reference pitch rate. Since successive controllers are tuned assuming the inner controllers are perfect, the simplified state space in Equation (4) can be taken further as shown in Equation (5), where q is the commanded reference pitch rate from the climb gradient controller.

$$\begin{bmatrix} \dot{W} \end{bmatrix} = \begin{bmatrix} -3.6757 & 39.7264 \end{bmatrix} \begin{bmatrix} W \\ q \end{bmatrix} \quad (5)$$

The climb gradient is then evaluated using Equation (6) where Θ is obtained from integrating the q state. The longitudinal velocity U is kept constant at equilibrium velocity U_e (41.1m/s) since it's assumed the airspeed controller will maintain this value.

$$\gamma = \Theta - \frac{W}{U_e} \quad (6)$$

Figure 9 shows the step response of the climb gradient controller against a unit step and a more realistic step of 3° . This value was chosen as it represents the climb gradient required to perform the most constraining climb gradient within the mission profile [3].

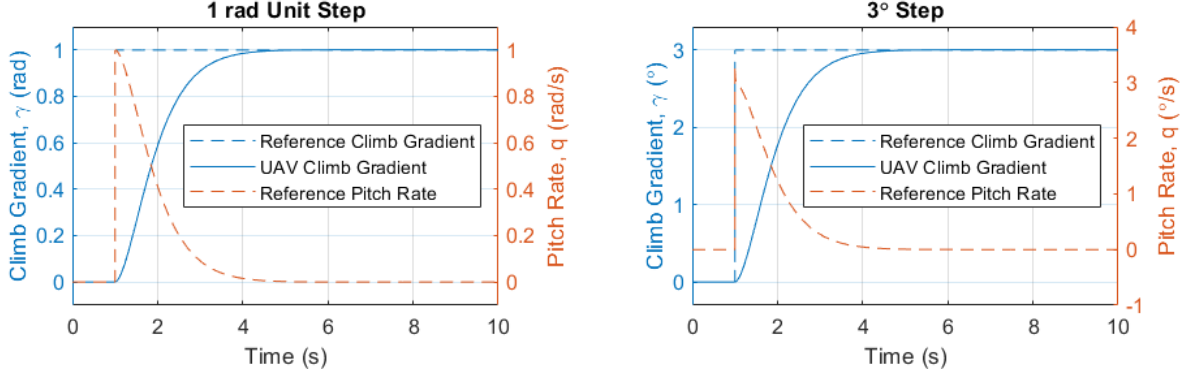


Figure 9. Climb Gradient Controller Step Reference Tracking

In tuning this controller, the rise time was intentionally kept higher than that of the pitch rate controller in order to avoid interference between the controllers. Table 7 shows the step response characteristics, illustrating the difference in rise times by roughly a factor of ten. The pitch rates commanded for the 3° step response are within reason, thus eliminating the need for a saturation limit on this controller.

Table 7. Climb Gradient Unit Step Response Characteristics and Controller Parameters

Rise Time (s)	Settle Time (s)	Gain Margin (dB)	Phase Margin ($^\circ$)	K_P	K_I	K_D	N
2.00	3.76	∞	97.2	1.00	0.01	0.01	100

5.3 Altitude Controller

The altitude controller is the outermost controller in the cascade loop, representing the final goal of this control structure. Altitude can be calculated from the climb gradient using Equation (7).

$$z = \int U_e \sin \gamma dt \quad (7)$$

Assuming a perfect climb gradient controller, Equation (7) represents the physical model used to tune the altitude controller. The step response of the tuned altitude controller can be seen in Figure 10 for a unit step and a more realistic step of 50m. The commanded climb gradient output was saturated at 3° representing the steepest climb phase of the mission profile. Permitting higher climb gradients negates the validity of the state space since the equations were linearised about zero climb gradient. The controller performance meets all requirements stipulated in Table 2 and the full step response characteristics and controller parameters can be observed in Table 8. As before the rise time is increased to avoid interference with the climb gradient controller. This is further reflected in the magnitude of the gains, which are significantly reduced from that of the climb gradient controller.

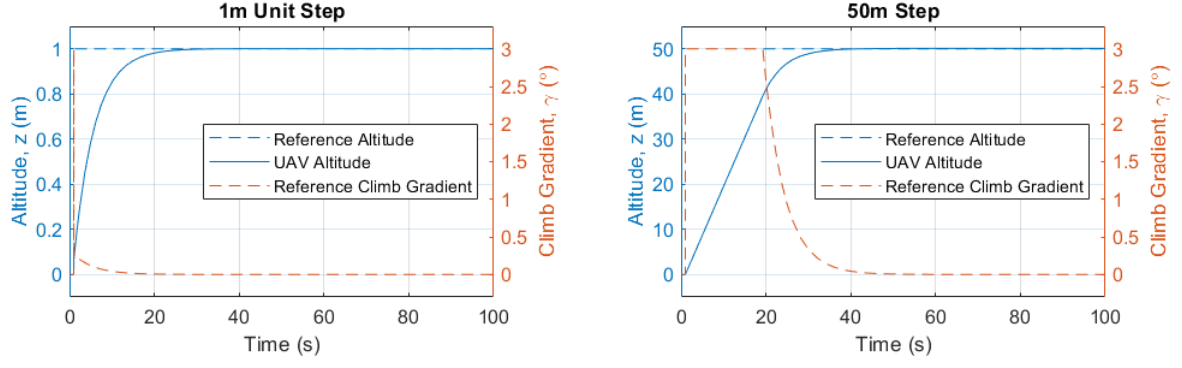


Figure 10. Altitude Controller Step Reference Tracking

Table 8. Altitude Unit Step Response Characteristics and Controller Parameters

Rise Time (s)	Settle Time (s)	Gain Margin (dB)	Phase Margin ($^{\circ}$)	K_P	K_I	K_D	N
10.82	26.6	∞	90.3	0.006	0.0005	0.005	100

6 Coupled Full State Verification

Following tuning of each of the individual controllers, it is important to verify the assumptions made in separating the states. The states were decoupled in order to facilitate the tuning process however, the controllers must be tested using the complete state space outlined in Equation (1) in order to verify their performance. Moreover, the tuning process for the triple cascaded controllers involved assuming perfect performance of the inner controllers. Therefore we must also verify the performance of the fully assembled cascaded controller to see if the stipulated criteria are still met. In order to carry this out, a step reference in airspeed at a constant altitude was given to test if altitude and pitch rate could in fact be maintained close to zero. Conversely a step reference in altitude was also given to see if equilibrium velocity could be maintained during climb. The results of this can be seen in Figures 11 and 12, carried out using the simlink architecture shown in Appendix C.

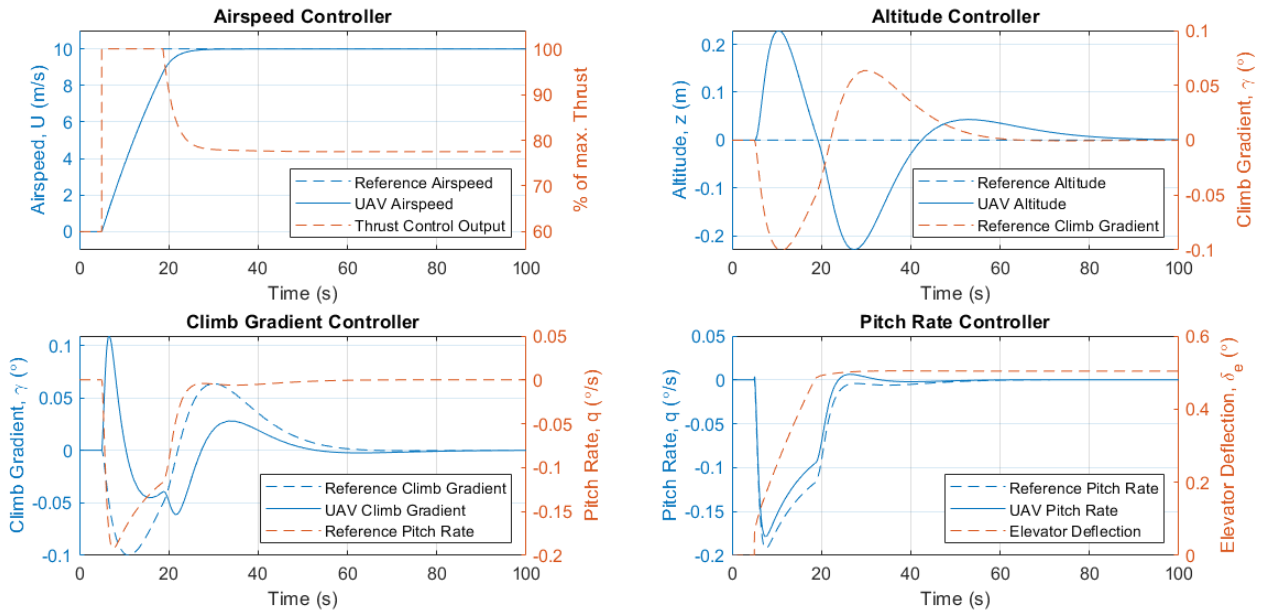


Figure 11. Full State Verification of Controllers for 10m/s Airspeed Step Response

Looking at Figure 11 we can see that consideration of all four states has minimal effect on the tracking of reference airspeed. The airspeed change induces perturbations in the other states, however the cascade controller appears to work well resulting a max. induced altitude change of 0.23m. This is well within the $\pm 30\text{ft}$ accuracy requirement stipulated in Table 2 thus validating the controller. Additionally the elevator deflection does not return to zero once the final airspeed value is reached. Initially this was a cause for concern, however it was later concluded that this elevator deflection likely remained to counteract the induced pitching moment from the increased thrust.

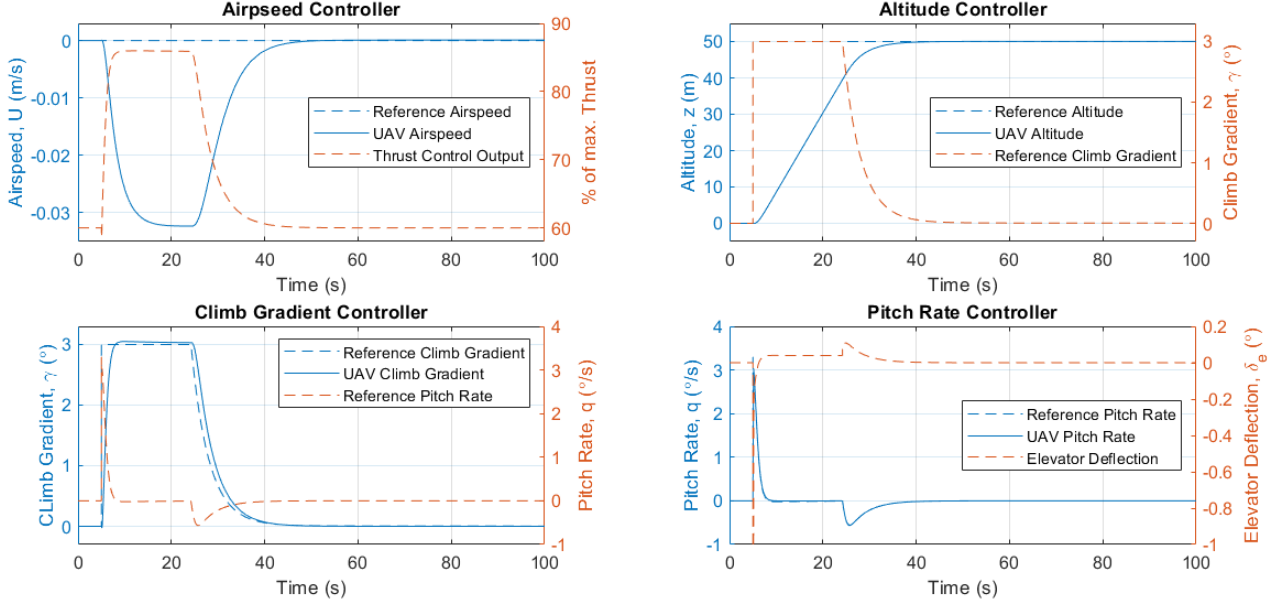


Figure 12. Full State Verification of Controllers for 50m Altitude Step Response

As with the airspeed step reference, Figure 12 shows minimal effect on altitude tracking with the introduction of all four states. The climb gradient and pitch rate controllers also show minimal deviation with respect to their reference inputs, thus validating our cascade tuning approach and the performance of the assembled cascade controller. A perturbation in the airspeed is observed due to coupling between climb gradient and the U state, however the magnitude was 0.032m/s which lies comfortably within our airspeed accuracy requirements outlined in Table 2.

7 Robustness Analysis

7.1 Wind Gusts

One of the criteria covered in section 2 stipulated wind gust resistance of up to magnitude 18m/s. Furthermore in section 3.1 the natural frequencies of the vibrational modes were identified, illustrating the frequencies of wind gusts likely to induce the largest perturbations. Consequently, the UAV was subjected to wind gusts at a range of frequencies surrounding the two identified natural frequencies in order to observe the maximum state perturbations. The wind gusts were modelled in both the x and z directions in order to capture both oscillatory modes, and their effect on the UAV was approximated through use of the U and W state derivatives (see Appendix A). Table 9 shows the results observed, both the open loop and closed loop responses were measured in order to verify controller effectiveness. The climb gradient and altitude controllers were not tested against wind gusts since it was assumed that the pitch rate controller will be designed to successfully eliminate perturbations.

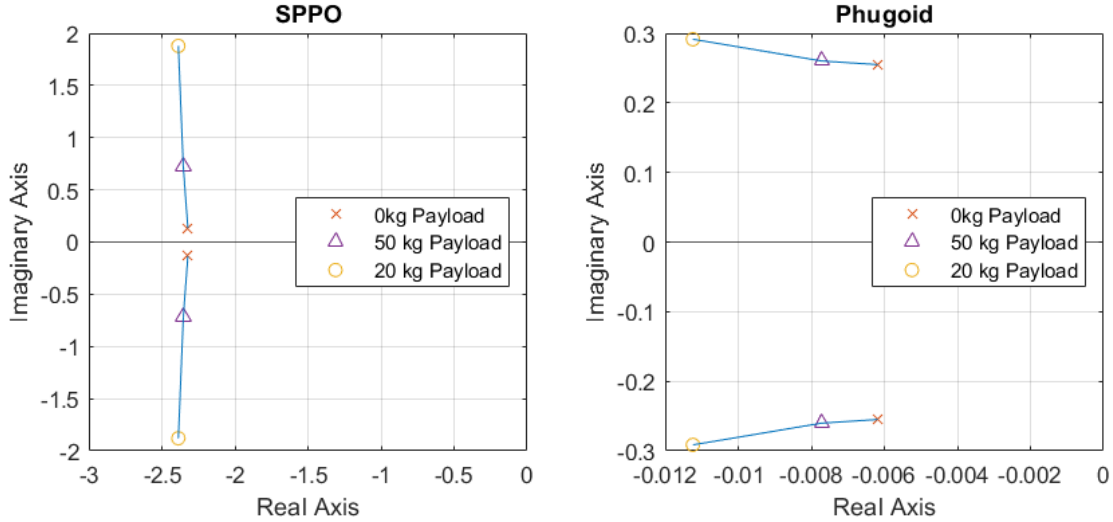
Table 9. Wind Gust Induced Max. Perturbations for the Open and Closed Loop System

Controller	State	Open Loop Max. Perturbation	Closed Loop Max. Perturbation
Airspeed	U	16m/s	0.06m/s
Pitch Rate	q	11.3°/s	1.1°/s

Looking at Table 9 we can see that for both the airspeed controller and pitch rate controller, perturbations are successfully attenuated when excited by the Phugoid and SPPO modes respectively. The closed loop max. perturbation in velocity lies comfortably within the airspeed accuracy requirements. As for pitch rate, due to the frequency of SPPO, the max. induced pitch angle was 1.4° which is less than our stipulated max climb gradient.

7.2 Payload and Fuel Mass Variation

The next stage of robustness analysis involves assessing the performance of our controller when payload and fuel mass are varied. From a control perspective this is important to consider, as varying masses can change the centre of gravity and static margin K_n . This can have a significant effect on the stability of the aircraft and performance of control surfaces. With this in mind, the the UAV longitudinal eigenvalues were plotted for all mass configurations outlined in Table 3, as can be seen in Figure 13. The origin was included in this plot to enable graphical observation of changes in natural frequency ω_n and damping ratio ζ .

**Figure 13.** Root Locus of Longitudinal UAV Eigenvalues for Different Payload and Fuel Masses

Looking at Figure 13 one of the first things to note is the justification behind choosing the 50kg payload configuration to form our initial state space. From a control perspective, this plot illustrates how this particular configuration is a 'middle ground' between all possible configurations, thereby making it the most ideal starting point.

Starting with SPPO, the 20kg payload configuration results in an increased natural frequency and reduced damping ratio, implying a more aggressive pitch rate controller might be needed. The converse is true for the 0kg payload configuration. The trend observed in this plot aligns with the standard trends for fixed wing aircraft [4] thereby providing validation for the state space development. On the other hand for Phugoid, the damping ratio appears to increase for the 20kg payload configuration, however as before, the natural frequency increases, implying that controllers for this configuration may have stricter rise time requirements.

To test our conclusions from the eigenvalue plots, the existing controllers were used with modified

state spaces representing each of the mass configurations. Each configuration was tested with a step airspeed reference whilst maintaining altitude, and also a step altitude reference whilst maintaining airspeed. The results of this can be seen in Figures 14 and 15.

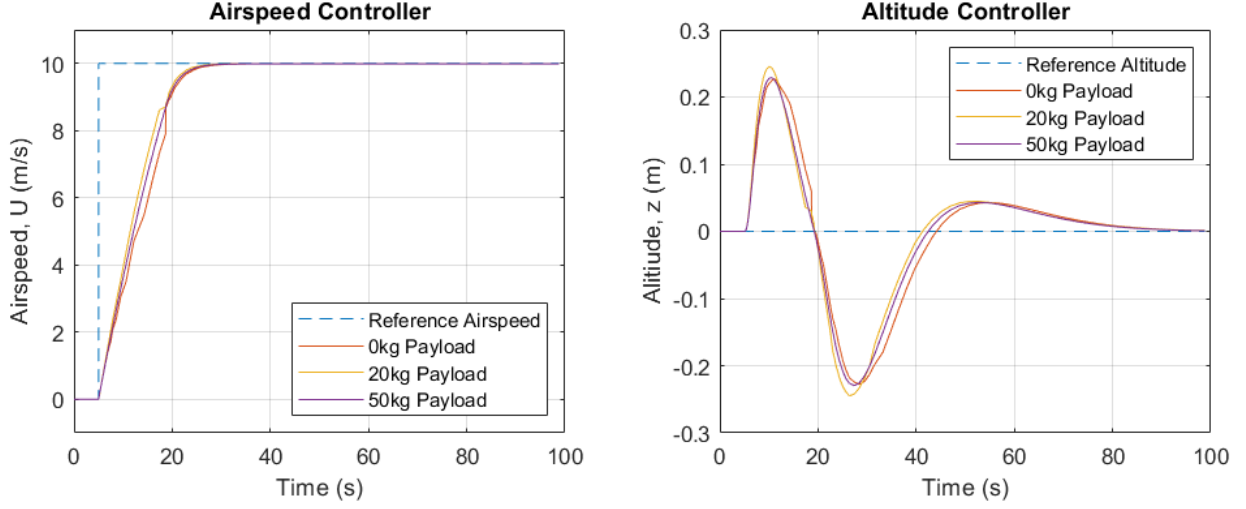


Figure 14. Effect of Mass Configuration on Control Performance for 10m/s Airspeed Step Response

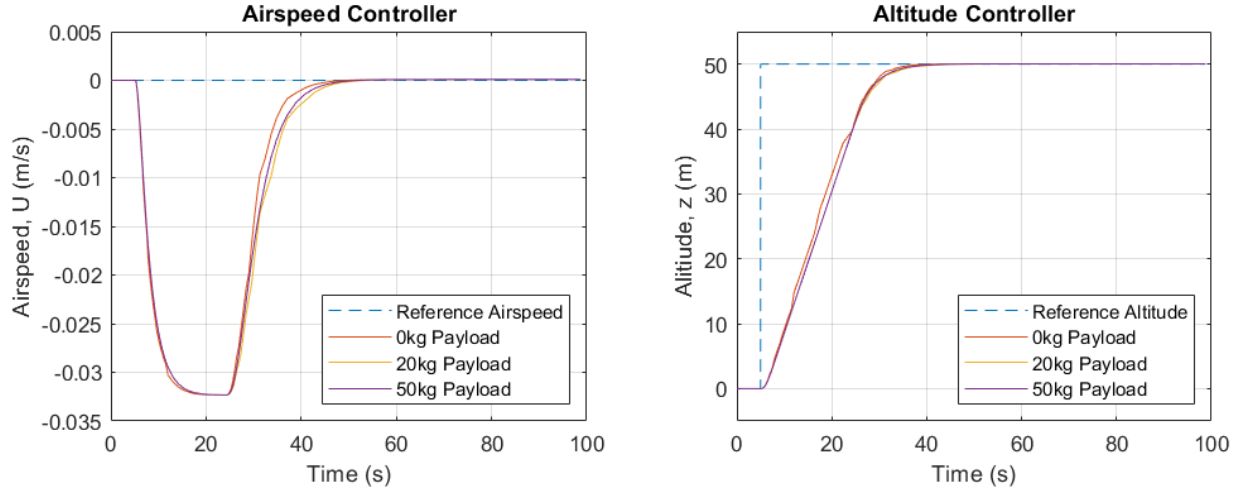


Figure 15. Effect of Mass Configuration on Control Performance for 50m Altitude Step Response

Looking at these figures, it is very apparent that the performance of our controllers is almost negligibly impacted by changes in mass configuration despite the potential issues outlined earlier. All control criteria outlined in Table 2 are still met, and thus we can conclude our robustness analysis.

8 Discussion

The final airspeed and altitude controllers indeed meet all the aims outlined in section 2. Despite simplifications made to the state space in tuning the individual controllers, and the assumptions made in tuning each of the loops in the cascade controllers, section 6 illustrates how when the assumptions are discarded, the controllers continue to perform satisfactorily. Moving forward, coupling between longitudinal and lateral states must also be considered. In development of the longitudinal state space, the Curie principle was employed, stipulating that the symmetric and asymmetric states are independent of each other [4]. While this is true to a reasonable extent, the coupling is not non-existent,

and must be modelled to test and ensure our controllers continue to track the given trajectories. Moreover when modelling the control outputs u_{δ_e} and u_T the only limits placed were that of maximum and minimum values. In reality there will be limits on the rates of change too. Elevator deflection for example will have a limit on the speed at which the angles can be changed. Figure 8 shows a full deflection and retraction of the elevator within 0.19s. While in theory possible, this is a very ambitious response. Looking ahead, in addition to saturation limits on the control outputs, limits should be placed on the rates at which these control outputs can vary to better reflect the physical response.

All controllers were validated by introducing coupling of all four symmetric states, and the state space was validated through comparison of the root locus plot in Figure 13 with that outlined in theory [4]. To provide further validation, the simulated response of the UAV could be compared to measured responses from similar UAVs to validate the physical response of changes in the control inputs.

Future development of the longitudinal fixed wing controller could include adaptation for other phases of the mission profile. Landing controllers could be developed as a back up in case the VTOL system fails, this would require perhaps a redesign of the control system about various other equilibria, and perhaps implementation of gain scheduling depending on current phase of flight.

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Appendix

A Longitudinal State Space Equations

The longitudinal state space of a fixed wing configuration aircraft can be assembled as outlined in Equation 8 [4].

$$\begin{bmatrix} m & -X_{\dot{W}} & 0 & 0 \\ 0 & m - Z_{\dot{W}} & 0 & 0 \\ 0 & -M_{\dot{W}} & I_{yy} & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \dot{U} \\ \dot{W} \\ \dot{q} \\ \dot{\Theta} \end{bmatrix} = \begin{bmatrix} X_U & X_W & X_q & -mg \cos \Theta \\ Z_U & Z_W & Z_q + mU_e & -mg \sin \Theta \\ M_U & M_W & M_q & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} U \\ W \\ q \\ \Theta \end{bmatrix} + \begin{bmatrix} X_{\delta_e} & X_{C_T} \\ Z_{\delta_e} & Z_{C_T} \\ M_{\delta_e} & M_{C_T} \\ 0 & 0 \end{bmatrix} \begin{bmatrix} \delta_e \\ C_T \end{bmatrix} \quad (8)$$

Where the stability derivatives $X_U = \frac{\partial X}{\partial U}$ are evaluated at the equilibrium point. The equations used to obtain each of the stability derivatives are shown below [4].

$$X_U = \frac{\partial T_x}{\partial V_\infty} - \rho U_e S_{ref} C_D \quad (9)$$

$$Z_U = \frac{\partial T_z}{\partial V_\infty} - \rho U_e S_{ref} C_L \quad (10)$$

$$M_U = l_T \frac{\partial T}{\partial V_\infty} \quad (11)$$

$$X_W = \frac{1}{U_e} \frac{\partial T_x}{\partial \alpha} + \frac{1}{2} \rho S_{ref} \left[-\frac{\partial C_D}{\partial \alpha} + C_L \right] \quad (12)$$

$$Z_W = -\frac{1}{U_e} \frac{\partial T_z}{\partial \alpha} - \frac{1}{2} \rho S_{ref} \left[\frac{\partial C_L}{\partial \alpha} + C_D \right] \quad (13)$$

$$M_W = \frac{1}{2} \rho S_{ref} \bar{c} \frac{\partial C_M}{\partial \alpha} \quad (14)$$

$$X_q = \frac{1}{2} \rho U_e S_{ref} l_H \left[-\frac{\partial C_{D_H}}{\partial \alpha} + C_{L_H} \right] \quad (15)$$

$$Z_q = -\frac{1}{2} \rho U_e S_{ref} l_H \left[\frac{\partial C_{L_H}}{\partial \alpha} + C_{D_H} \right] \quad (16)$$

$$M_q = l_H^2 (Z_W)_H - h_H l_H (X_W)_H + l_H (M_W)_H \quad (17)$$

$$X_{\dot{W}} = \frac{l_H}{U_e} \frac{d\epsilon}{d\alpha} (X_W)_H \quad (18)$$

$$Z_{\dot{W}} = \frac{l_H}{U_e} \frac{d\epsilon}{d\alpha} (Z_W)_H \quad (19)$$

$$M_{\dot{W}} = \frac{l_H}{U_e} \frac{d\epsilon}{d\alpha} [-h_H (X_W)_H + l_H (Z_W)_H + (M_W)_H] \quad (20)$$

$$X_{\delta_e} = -\frac{1}{2} \rho U_e^2 S_H d_E \quad (21)$$

$$Z_{\delta_e} = -\frac{1}{2} \rho U_e^2 S_H a_E \quad (22)$$

$$M_{\delta_e} = l_H Z_{\delta_e} - h_H X_{\delta_e} + \frac{1}{2} \rho U_e^2 S_H \bar{c}_H \frac{dC_{M_H}}{d\delta_e} \quad (23)$$

$$X_{C_T} = \frac{1}{2} \rho U_e^2 S_{ref} \cos \alpha \quad (24)$$

$$Z_{C_T} = -\frac{1}{2} \rho U_e^2 S_{ref} \sin \alpha \quad (25)$$

$$M_{C_T} = \frac{1}{2} \rho U_e^2 S_{ref} l_T \quad (26)$$

B Tuning Environment

A typical tuning environment can be seen here demonstrating use of pole placement, gain/phase margins and step response to design the controllers [6].

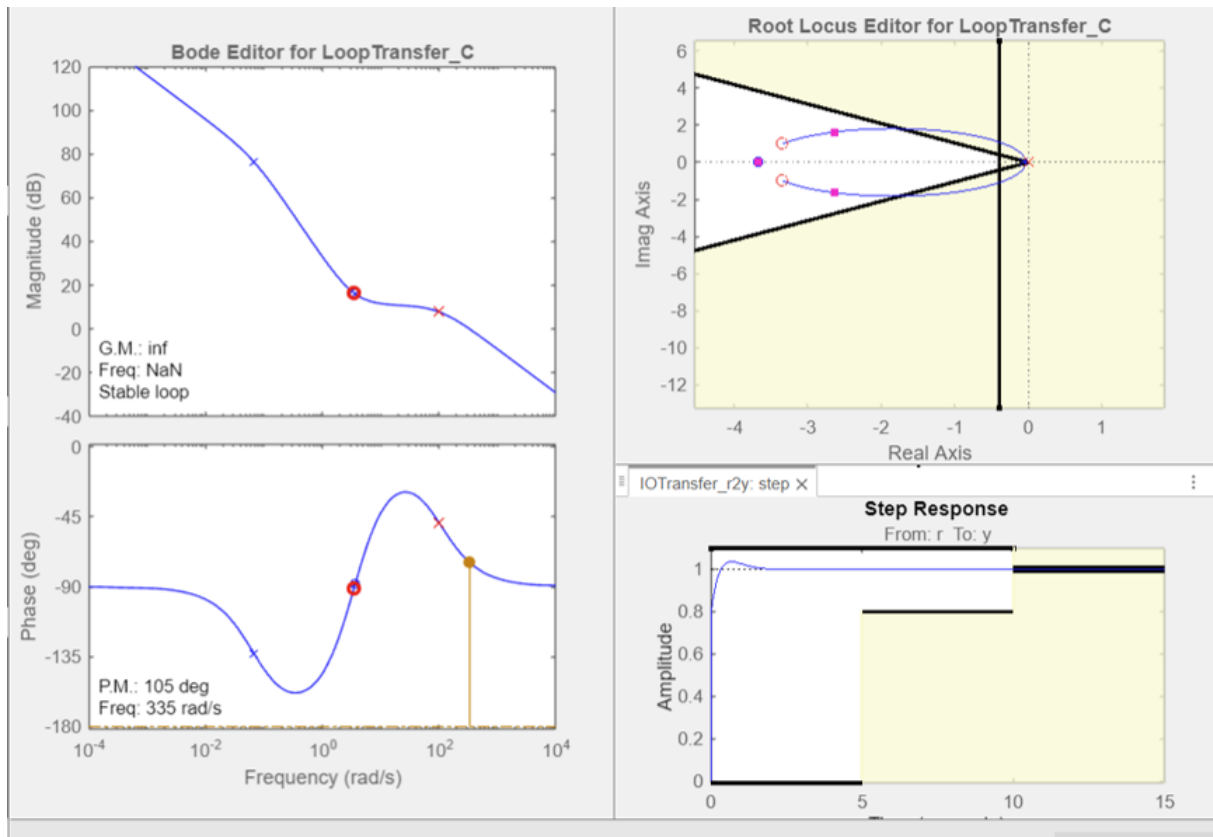


Figure 16. MATLAB Control System Designer Tuning Environment

C Simulink Architecture

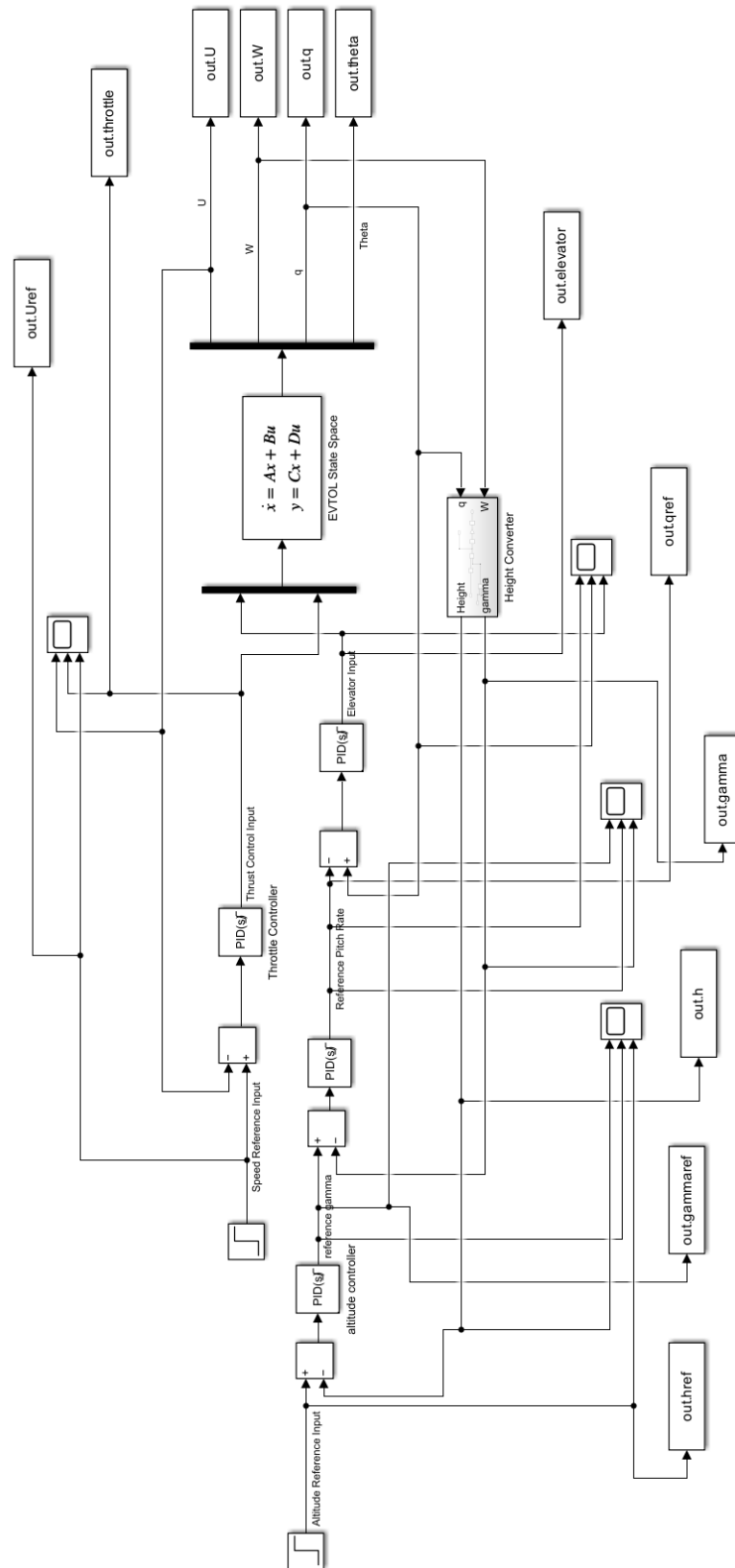


Figure 17. Complete Longitudinal State Space Simulink Block Diagram